HypRQ-VAE: Hyperbolic Item Indexing for Long-Tail-Aware Generative Recommender Systems

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***Abstract*—Sequential recommender systems model user behavior as item-ID sequences, while recent generative methods cast recommendation as a language modeling task using large language models (LLMs). While this paradigm incorporates rich textual semantics, it introduces a fundamental mismatch: LLMs operate on text tokens, whereas recommender systems depend on discrete item indices. This misalignment often leads to hallucinations in generative recommendations. Existing methods attempt to bridge this gap by learning item vocabularies in Euclidean space, but they struggle to model the inherent long-tail distribution of real-world catalogs, where a small number of head items dominate, and a vast number of tail items reflect users' niche preferences. To address this issue, we introduce Hyperbolic Residual-Quantized Variational AutoEncoder (HypRQ-VAE), the first framework to learn item indexing in hyperbolic space. HypRQ-VAE leverages the unique properties of hyperbolic geometry, whose exponential volume expansion naturally accommodates the power-law structure of user-item interactions. This allows the model to encode rich textual semantics while preserving the representational fidelity of sparse, long-tail items. Experiments on three benchmark datasets show that HypRQ-VAE significantly improves the performance of recommendation, particularly in recommending tail items. Our analysis attributes these gains to the superior capacity of hyperbolic space to model item hierarchies and sparsity in generative recommendation. Our code and data are available at: https://github.com/wulongfeng/HypRQ-VAE.**

***Index Terms*—Generative Recommendation, Semantic IDs, Hyperbolic Representation**

I. Introduction

In today's data-saturated digital world, recommender systems (RS) play a crucial role in alleviating information overload by delivering personalized and relevant content, which streamlines users' decision-making and boosts service efficiency. Since user preferences dynamically evolve over time, sequential recommendation has attracted significant attention for its ability to capture the temporal patterns in user behavior. Most state-of-the-art models are built on sequentially created interaction logs where each event is encoded by an item ID [1, 2]. To model these sequences, researchers have explored a wide range of deep learning architectures, such as GNNs [3] and Transformers [4], and enriched the basic item-ID signal with auxiliary content such as titles, descriptions, and categories. More recently, pre-trained language models (PLMs) have been introduced to better exploit the textual semantics in item metadata, further improving recommendation performance [5–7].

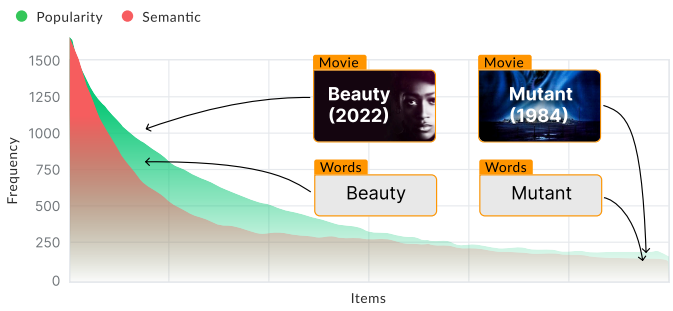


Fig. 1. An illustrative figure of long-tail distribution in item popularity and semantic tokens.

The remarkable success of Large Language Models (LLMs) across diverse language tasks [8–11] has inspired their application to RS by framing recommendation as a generation problem [12, 13]. However, a fundamental challenge persists: the semantic misalignment between the language representations that LLMs excel at processing and the collaborative filtering signals foundational to RS. Traditional recommenders model user behavior as sequences of discrete item IDs, while LLMs operate on textual tokens. This vocabulary mismatch hinders the effective use of LLMs for generating accurate recommendations. Although items often come with textual information through titles or descriptions, these texts often comprise tens to thousands of tokens, making it exceedingly difficult for LLMs to generate precise, real-world item recommendations without hallucination [14]. Conversely, using

raw item IDs (e.g., '01234') leads to an unmanageably large, semantically meaningless vocabulary (e.g., <012><34>) [12], which fails to capture relationships among items. To bridge this gap, recent works have proposed specialized item-indexing mechanisms that construct item vocabularies and train LLMs to generate target items directly [15, 16]. A notable example is the Residual-Quantized Variational AutoEncoder (RQ-VAE) [17], a multi-level vector quantizer that recursively quantizes the residual vectors, from coarse to fine, to generate a set of codewords, producing item indices that capture the hierarchical structure of items. However, these methods typically operate in Euclidean space, which struggles with the long-tail distribution prevalent in real-world recommendation data.

Large-scale recommendation datasets commonly exhibit a pronounced long-tailed distribution [18]: a small fraction of popular "head" items dominate user interactions, while a vast number of "tail" items reflect niche interests and emerging trends. Both of them are essential for a well-rounded recommender system. Meanwhile, item semantics also follow the long-tailed distribution, as illustrated in Figure 1. With these two distributions sharing implicit correlations, popular items typically exhibit more generic semantic content, while rare items display greater semantic distinctiveness. However, traditional Euclidean methods disproportionately emphasize head items while neglecting tail items [19]. To better model this inherent imbalance, researchers have turned to hyperbolic space, whose exponentially expanding volume naturally accommodates the power-law structure of user-item interactions. Although hyperbolic recommendation models have shown promising empirical results [19, 20], two key questions remain unsolved: *(1) How effective are hyperbolic models at generating item IDs?* and *(2) Do hyperbolic models outperform Euclidean approaches, especially for tail-items recommendation?*

To answer the above questions, we introduce Hyperbolic RQ-VAE (HypRQ-VAE), a novel framework that generates semantic IDs for items in hyperbolic space. Our goal is to leverage hyperbolic geometry's exponentially increased capacity and its inherent ability for modeling hierarchical structures to provide greater representational flexibility, particularly for rare semantic concepts. We conduct a comparative analysis between hyperbolic and Euclidean models, partitioning items into the top 20% "head" (H20) and bottom 80% "tail" (T80) based on the Pareto Principle [1]. Our experiments reveal that hyperbolic models allocate substantially more representational capacity to tail items than Euclidean models (see Section IV-C). We argue that hyperbolic geometry, with its high representational capacity and its natural fit for hierarchical data, can yield richer and more discriminative representations for tail items than Euclidean embeddings. While prior research has incorporated various signals for item indexing, to the best of our knowledge, this is the first work to incorporate hyperbolic representation into the item indexing process for generative recommendation. Our main contributions are summarized as follows:

- **Problem.** We are the first to explore the problem of item indexing in the hyperbolic space for generative recommendation tasks, addressing the limitations of Euclidean geometry in capturing hierarchical item relationships.
- **Model.** We introduce HypRQ-VAE, a novel indexing framework that integrates rich item semantics with a hyperbolic geometry naturally suited to the long-tailed distribution characteristic of recommender systems.
- **Evaluation.** We conduct extensive experiments on three widely adopted datasets (Instruments, Arts from the Amazon, and MovieLens) with standard sequential models and competing indexing strategies, demonstrating the effectiveness of our approach, especially on tail items. Our in-depth analysis highlights how hyperbolic representations contribute to these improvements.

[1]https://en.wikipedia.org/wiki/Pareto_principle

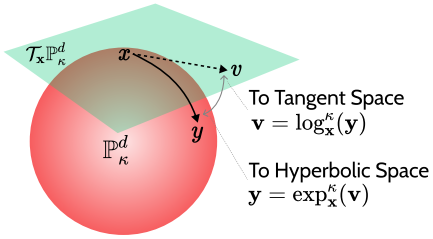


Fig. 2. Illustration of the Poincaré ball model and its associated exponential and logarithmic maps.

II. Preliminaries

Riemannian Geometry. A d-dimensional Riemannian manifold $\mathcal{M}$, denoted as $(\mathcal{M}, \mathfrak{g})$, is a topological space equipped with a metric tensor $\mathfrak{g}$ [21]. At any point $\mathbf{x}$ in $\mathcal{M}$, the manifold can be locally approximated by a local linear space called the tangent space $\mathcal{T}_{\mathbf{x}}\mathcal{M}$, which is isometric to $\mathbb{R}^d$. The geodesic represents the shortest path between two points on the manifold [22], and its length defines the induced distance $d_{\mathcal{M}}$. Notably, in hyperbolic space, the hyperbolic distance to the origin (HDO) serves as the induced hyperbolic norm. The exponential map $\exp_{\mathbf{x}}$ projects tangent vectors onto the manifold, and its reverse function, the logarithmic map $\log_{\mathbf{x}}$, returns points to the tangent space. Furthermore, parallel transport $P_{\mathbf{x}\to\mathbf{y}}$ enables the transportation of geometric data from $\mathbf{x}$ to $\mathbf{y}$ along the unique geodesics while preserving the metric tensor $\mathfrak{g}$.

Hyperbolic Space. Hyperbolic space refers to a Riemannian manifold with a constant negative curvature, and its coordinates can be represented using various isometric models [23, 24]. In the domain of machine learning and deep learning, the Poincaré ball model has emerged as a particularly valuable framework, offering unique advantages for embedding hierarchical structures and complex relational data in a bounded continuous space [25–28]. A d-dimensional Poincaré ball model with a constant negative curvature $\kappa(\kappa < 0)$ is defined as a Riemannian manifold $(\mathbb{P}^d_\kappa, \mathfrak{g}^\kappa)$, where $\mathbb{P}^d_\kappa = \{x \in \mathbb{R}^d | \|\mathbf{x}\|^2 < -\frac{1}{\kappa}\}$ and its metric tensor $\mathfrak{g}^\kappa = (\lambda^\kappa_{\mathbf{x}})^2\mathbf{I}_d$ and $\mathbf{I}_d$ is d-dimensional identity matrix. Here, $\mathbb{P}^d_\kappa$ is an d-dimensional open ball of radius $(-\kappa)^{-\frac{1}{2}}$, and $\lambda^\kappa_{\mathbf{x}} = 2(1 + \kappa||\mathbf{x}||^2)^{-1}$ is the conformal

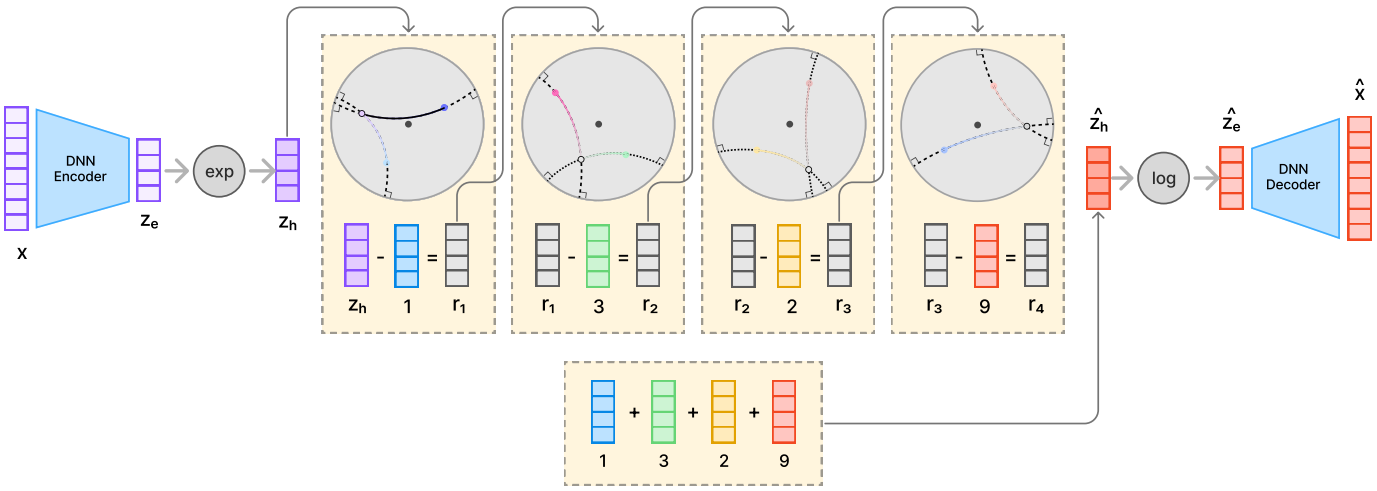


Fig. 3. HypRQ-VAE: The figure depicts the encoding process for an item. An item embedding $\mathbf{x}$ is passed through a DNN encoder to obtain a latent vector z_e. This vector is then projected into hyperbolic space via an exponential map to produce z_h, which serves as the initial residual r_0 (purple bar). This residual is then processed sequentially through multiple quantization levels. At each level i, the quantizer identifies the nearest code vector $e_{c_i}^{l_i}$ from its codebook and computes the next residual $r_{i+1} = r_i \ominus_\kappa e_{c_i}^{l_i}$ using Möbius subtraction. For instance, at the first level, the code vector $e_{c_1}^{l_1}$ (blue bar) is selected, and at the second level, $e_{c_2}^{l_2}$ (green bar) is chosen. The final semantic code is the sequence of indices of these selected vectors, exemplified by $(1, 3, 2, 9)$.

factor. This induces the inner product $\langle \mathbf{p}, \mathbf{q} \rangle_{\mathbf{x}}^{\kappa} = (\lambda_{\mathbf{x}}^{\kappa})^2 \langle \mathbf{p}, \mathbf{q} \rangle$ and norm $||\mathbf{p}||_{\mathbf{x}}^{\kappa} = \lambda_{\mathbf{x}}^{\kappa} ||\mathbf{p}||$ for $\mathbf{p}, \mathbf{q} \in \mathcal{T}_{\mathbf{x}} \mathbb{P}_{\kappa}^{d}$. The exponential and logarithmic maps, $\exp_{\mathbf{x}}^{\kappa}$ and $\log_{\mathbf{x}}^{\kappa}$, are defined below.

Exponential and Logarithmic Map. This framework defines several operations, including Möbius addition $\oplus_\kappa$, distance between points on the manifold $d_\kappa(\mathbf{x}, \mathbf{y})$, exponential and logarithmic maps. Given their relevance to our work, we present the definitions of the exponential and logarithmic maps below. These operations are defined within the Riemannian manifold framework, specifically considering the Poincaré ball model $\mathbb{P}_{\kappa}^{d}$:

$$
\begin{aligned}
\mathbf{p} \oplus_\kappa \mathbf{q} &= \frac{(1 - 2\kappa\langle \mathbf{p}, \mathbf{q} \rangle - \kappa ||\mathbf{q}||^2)\mathbf{p} + (1 + \kappa ||\mathbf{p}||^2)\mathbf{q}}{1 - 2\kappa\langle \mathbf{p}, \mathbf{q} \rangle + \kappa^2 ||\mathbf{p}||^2 ||\mathbf{q}||^2}, \\
\exp_{\mathbf{x}}^{\kappa}(\mathbf{v}) &= x \oplus_\kappa (\tanh(\sqrt{-\kappa}\frac{\lambda_{\mathbf{x}}^{\kappa} ||\mathbf{v}||}{2}) \frac{\mathbf{v}}{\sqrt{-\kappa}||\mathbf{v}||}), \\
\log_{\mathbf{x}}^{\kappa}(\mathbf{v}) &= \frac{2}{\sqrt{-\kappa}\lambda_{\mathbf{x}}^{\kappa}} (\tanh^{-1}(\sqrt{-\kappa}|| -\mathbf{x} \oplus_\kappa \mathbf{v}||) \frac{-\mathbf{x} \oplus_\kappa \mathbf{v}}{|| -\mathbf{x} \oplus_\kappa \mathbf{v}||}),
\end{aligned}
\tag{1}
$$

The induced Möbius subtraction $\ominus_\kappa$ is given by $\mathbf{p} \ominus_\kappa \mathbf{q} = \mathbf{p} \oplus_\kappa (-\mathbf{q})$. And the exponential and logarithmic maps enable transitions between the tangent space and the hyperbolic space (as shown in Figure 2).

III. Methodology

In this section, we provide a detailed description of our proposed HypRQ-VAE framework. We begin with a high-level overview of the architectural pipeline before detailing the two primary stages of our methodology: (1) semantic ID generation via hyperbolic residual quantization, and (2) a generative recommender that leverages these structured identifiers to bridge the gap between language modeling and recommender systems. By situating the indexing process within a hyperbolic manifold, our approach specifically addresses the representational challenges inherent in long-tail data distributions.

A. Overview of the Approach

In Section I, we discussed a key limitation of using LLMs for recommendation: the mismatch between the textual semantics captured by LLMs and the collaborative semantics essential for effective recommendation. To address this fundamental limitation, we propose a two-stage strategy that enhances semantic integration.

Generation of Semantic IDs. We use hyperbolic RQ-VAE to compress the textual embedding of each item into several learned discrete IDs. Specifically, we project the representation of items from tangent space into hyperbolic space via exponential and logarithmic maps, enabling the learned item indices to better capture similarities between the textual semantics of items while providing unique indexing representations for specific items. Furthermore, hyperbolic geometry's exponential volume growth and natural support for hierarchical structure further boost representational capacity, especially for long-tail items, which tend to receive less attention in conventional Euclidean models.

Generative Recommender with Semantic IDs. To bridge the semantic gap, we introduce a series of prompt instructions aligning textual semantics and collaborative semantics. By fine-tuning on sequences of semantic IDs, we integrate collaborative knowledge into the LLM's generative process, enabling it to make more effective recommendations.

B. Generation of Semantic IDs

A key challenge in adapting LLMs for recommendation is encoding items into a compact set of discrete tokens. We

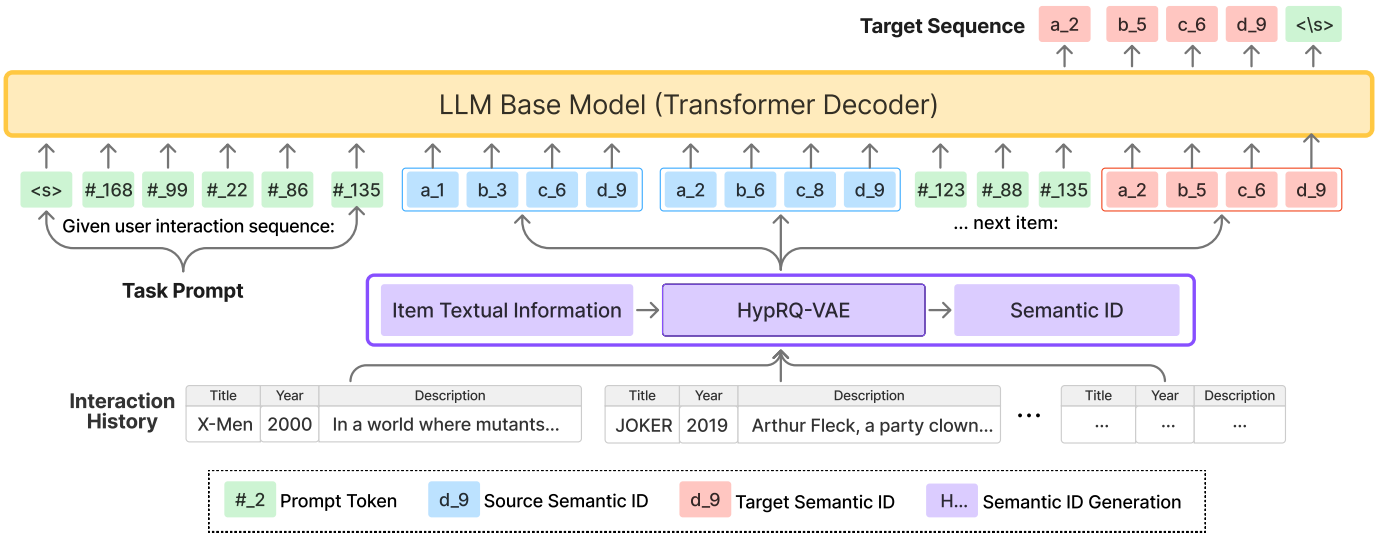


Fig. 4. Overview of our Transformer decoder-based architecture for generative recommendation using semantic IDs. The figure shows the Supervised Fine-Tuning (SFT) stage. A causal attention mask ensures autoregressive learning by forcing the model to predict next token based only on preceding tokens.

address this by generating a Semantic ID for each item in the hyperbolic space, a structured sequence of learned indices derived from its textual features. Our method is built upon RQ-VAE. We reformulate the quantization in hyperbolic space, whose geometric properties are better suited for modeling the hierarchical and long-tail nature of item data.

Our model first uses an encoder to map an item's embedding $\mathbf{x}$ into a Euclidean latent representation $z_e \in \mathbb{R}^d$. This vector is then projected onto the Poincaré ball $\mathbb{P}_\kappa^d$ using the exponential map $\exp_0^\kappa$, yielding the projected latent representation $z_h \in \mathbb{P}_\kappa^d$. This vector z_h serves as the initial residual, r_0, for the quantization process. At each quantization level l, we have a codebook $\mathcal{C}^l = \{e_k^l\}_{k=1}^K$, where K is the dimension of the codebook, and each code vector e_k^l is a learnable cluster center. The residual quantization process is performed recursively within the hyperbolic space. At each level, we identify the index c_i of the closest code vector and compute the residual for the next level, r_{l+1}, using Möbius subtraction:

$$c_i = \arg\min_k ||r_i \ominus_\kappa e_k^i||_2^2, \qquad r_{i+1} = r_i \ominus_\kappa e_{c_i}^i. \tag{2}$$

The final quantized representation $\hat{z_h}$ is constructed by aggregating the selected code vectors using Möbius addition: $\hat{z}_h = e_{c_1}^1 \oplus_\kappa \cdots \oplus_\kappa e_{c_L}^L$. This hyperbolic vector is then projected back to Euclidean space with the logarithmic map $\log_0^\kappa$ before being passed to the decoder to reconstruct the original embedding $\hat{\mathbf{x}}$.

The model is trained by minimizing the loss function below, $\mathcal{L}_{\text{RQ-VAE}}$:

$$\begin{aligned} \mathcal{L}_{\text{RECON}} &= ||\mathbf{x} - \hat{\mathbf{x}}||_2^2, \\ \mathcal{L}_{\text{RQ}} &= \sum_{i=1}^{L} ||\text{sg}[r_i] - e_{c_i}^i||_2^2 + \beta||r_i - \text{sg}[e_{c_i}^i]||_2^2, \\ \mathcal{L}_{\text{RQ-VAE}} &= \mathcal{L}_{\text{RECON}} + \gamma\mathcal{L}_{\text{RQ}}, \end{aligned} \tag{3}$$

where $\mathcal{L}_{\text{RECON}}$ is the reconstruction loss and $\mathcal{L}_{\text{RQ}}$ is the quantization loss that optimizes the codebook embeddings. The term $\text{sg}[*]$ denotes the stop-gradient operator, while β and γ are weighting coefficients. The sequence of indices $(c_1, \cdots, c_L)$ generated during quantization forms the core of the Semantic ID. As shown in Figure 3, an index sequence like (1, 3, 2, 9) is formatted into the semantic ID of this item as <a_1><b_3><c_2><d_9>.

Unlike conventional vector quantization, RQ provides significantly larger representational capacity with a smaller codebook. Moreover, its coarse-to-fine quantization naturally produces a hierarchical item index, which aligns naturally with autoregressive generation. When extended to hyperbolic space, these benefits are amplified: the exponential expansion of hyperbolic space allows more expressive and hierarchically structured embedding, enhancing representation and understanding of the diverse items, including both head and long-tail items.

Handling Collisions. Compressing item content into discrete semantic IDs can cause collisions, where different items are mapped to the same tokens. Prior works mitigated this issue by appending auxiliary identifiers for conflicting items [15, 29], which can introduce semantically unrelated information. Specifically, LC-Rec [16] employs a uniform distribution constraint in the final layer to minimize item indexing conflicts, while LETTER [30] introduces regularizers to enhance code assignment diversity. However, these methods do not completely resolve collisions, especially when the number of colliding items exceeds the capacity of the last-level codebook or when semantically similar items are inherently difficult to distinguish.

To address this challenge, we utilize a systematic token reassignment methodology based on codebook proximity metrics. Given a set of M colliding items, we construct a distance tensor $\mathbf{D} \in \mathbb{R}^{M \times L \times K}$, where each element $d_k^i \in \mathbf{D}$

TABLE I
COMPARISON OF METHODS ACROSS THREE DATASETS. H AND N DENOTE HIT RATE AND NDCG, RESPECTIVELY. BOLD NUMBERS INDICATE THE BEST RESULTS, WHILE NUMBERS WITH AN UNDERLINE INDICATE THE SECOND-BEST. THE FINAL ROW DEPICTS THE % IMPROVEMENT WITH OUR METHOD OVER THE STRONGEST BASELINE.

Method	Movie Lens				Instruments				Arts			
	H@5	H@10	N@5	N@10	H@5	H@10	N@5	N@10	H@5	H@10	N@5	N@10
MF	0.0343	0.0609	0.0220	0.0305	0.0444	0.0551	0.0364	0.0399	0.0202	0.0262	0.0162	0.0181
Caser	0.0422	0.0651	0.0276	0.0349	0.0525	0.0706	0.0479	0.0541	0.0267	0.0394	0.0191	0.0232
SASRec	0.0518	0.0936	0.0274	0.0435	0.0596	0.0717	0.0336	0.0405	0.0381	0.0519	0.0212	0.0269
P5-TID	0.0228	0.0281	0.0157	0.0175	0.0002	0.0002	0.0001	0.0001	0.0006	0.0006	0.0004	0.0005
P5-CID	0.0612	0.0920	0.0398	0.0503	0.0537	0.0626	0.0470	0.0498	0.0400	0.0498	0.0324	0.0355
TIGER	0.0565	0.0856	0.0386	0.0480	0.0608	0.0716	0.0529	0.0563	0.0412	0.0506	0.0337	0.0367
LC-Rec	0.0553	0.0871	0.0355	0.0456	0.0620	0.0780	0.0530	0.0570	0.0413	0.0505	0.0340	0.0369
LETTER	0.0649	0.0964	0.0410	0.0535	0.0600	0.0730	0.0520	0.0550	0.0383	0.0495	0.0317	0.0353
Ours	**0.0652**	**0.1010**	**0.0423**	**0.0539**	**0.0690**	**0.0830**	**0.0600**	**0.0650**	**0.0425**	**0.0528**	**0.0348**	**0.0381**
	+ 0.5%	+ 4.8%	+ 3.2%	+ 0.7%	+ 11.3%	+ 6.4%	+ 13.2%	+ 14.0%	+ 2.9%	+ 4.3%	+ 2.4%	+ 3.3%

represents the hyperbolic distance between the residual vector r_i and code vector e_k^i across all codebook levels. After sorting these distances along the code dimension to generate an index $\mathbf{I}$. We use a cascading assignment strategy starting from the last level. Each item is first assigned its nearest available token. When multiple items compete for the same token, the item with the smallest distance is assigned that token, while the others move to their next closest available choices. If the last level cannot provide unique tokens for all items, the same procedure is applied to the preceding level and repeated upward until every item receives a unique identifier.

C. Generative Recommender with Semantic IDs

We formulate next-item prediction as a sequence-to-sequence generation task over semantic IDs. As illustrated in Figure 4, a HypRQ-VAE is first trained on the textual content of items in the user's Interaction History to encode each item into a discrete, l-length Semantic ID $(c_{i,1}, \cdots, c_{i,l})$. A user's history is then represented as a chronologically ordered sequence of these IDs. This sequence is flattened into a single token stream: $(c_{1,1}, \cdots, c_{1,l}, c_{2,1}, \cdots, c_{2,l}, \cdots, c_{n,1}, \cdots, c_{n,l})$, and subsequently embedded within a prompt template to form the full input sequence (see the prompt template in Section IV for an example).

The model is trained autoregressively to generate the ID of the next item, $(c_{n+1,1}, \cdots, c_{n+1,l})$. We optimize the model by minimizing the negative log-likelihood of the target sequence:

$$\mathcal{L}_\theta = -\sum_{j=1}^{|\mathbf{Y}|} \log P_\theta(\mathbf{Y}_j | \mathbf{Y}_{<j}, \mathbf{X}), \tag{4}$$

where θ demotes the model parameters, $\mathbf{X}$ is the input prompt containing historical item IDs, $\mathbf{Y}$ is the target item's ID sequence, and $\mathbf{Y}_{<j}$ are all tokens in the target sequence before the j-th token.

During inference, the goal is to generate the next item that best aligns with the user's preferences. The model autoregressively produces the sequence of semantic ID tokens: $\hat{\mathbf{Y}}_t = \arg\max_{v \in \mathcal{V}} P_\theta(v | \hat{\mathbf{Y}}_{<t}, \mathbf{X})$, where $\mathcal{V}$ is the vocabulary including all possible semantic ID tokens.

IV. EXPERIMENTS

In this section, we conduct a comprehensive evaluation of HypRQ-VAE for generative recommendation. We first describe the experimental setup, including datasets, baselines, and implementation details. We then report the main results along with in-depth ablation studies and representational analyses that further examine the model's robustness and effectiveness, particularly for long-tail recommendation.

A. Experimental Settings

TABLE II
STATISTICS OF THREE REAL-WORLD BENCHMARKS.

Datasets	# Users	# Items			# Interactions	# Density
		All	H20(%)	T80(%)		
Movie Lens	6,040	2,751	59.97	40.03	1,000,000	6.018%
Instruments	24,773	9,922	61.00	39.00	206,153	0.084%
Arts	45,142	20,956	57.65	42.35	390,832	0.041%

Datasets. We evaluate our model on three widely adopted real-world recommendation datasets: MovieLens [2], and the Instruments and Arts from the Amazon [3] dataset [31]. Each item includes both a title and a description; for MovieLens, we augment each movie with its plot overview from TMDB[4] as the description of it. Following previous work [32], we filter out users and items with fewer than five interactions, then order each user's remaining interactions chronologically to form behavior sequences. To align with baseline settings, we truncate all sequences to a maximum length of 20. The statistics of the dataset are in Table II, where H20 and T80 denote the average ratio of the head items and tail items appearing in the user's preference.

[2]https://movielens.org/
[3]https://jmcauley.ucsd.edu/data/amazon/
[4]https://www.themoviedb.org/

TABLE III
PERFORMANCE COMPARISON OF EUCLIDEAN AND HYPERBOLIC METHODS ON THE THREE DATASETS. THE RELATIVE IMPROVEMENTS COMPARED WITH EUCLIDEAN METHOD IS DENOTED BY $\Delta_{\mathcal{H}}$.

Metric	Method	Movie Lens			Instruments			Arts		
		H20	T80	All	H20	T80	All	H20	T80	All
Hit@5	TIGER	0.0872	0.0103	0.0565	0.0982	0.0013	0.0608	0.0691	0.0031	0.0412
	Ours	0.0914	0.0153	0.0652	0.1099	0.0018	0.0690	0.0707	0.0036	0.0425
	$\Delta_{\mathcal{H}}$	+ 4.82%	+ 48.54%	+ 15.52%	+ 11.88%	+ 31.95%	+ 13.42%	+ 2.32%	+ 16.13%	+ 3.29%
Hit@10	TIGER	0.1292	0.0203	0.0856	0.1161	0.0021	0.0716	0.0834	0.0058	0.0506
	Ours	0.1474	0.0310	0.1010	0.1305	0.0028	0.0830	0.0868	0.0068	0.0528
	$\Delta_{\mathcal{H}}$	+ 14.09%	+ 52.71%	+ 17.97%	+ 12.34%	+ 36.23%	+ 15.97%	+ 4.08%	+ 17.24%	+ 4.35%
NDCG@5	TIGER	0.0578	0.0067	0.0386	0.0865	0.0008	0.0529	0.0570	0.0020	0.0337
	Ours	0.0599	0.0092	0.0423	0.0954	0.0010	0.0600	0.0583	0.0024	0.0348
	$\Delta_{\mathcal{H}}$	+ 3.63%	+ 37.31%	+ 9.62%	+ 10.23%	+ 23.99%	+ 13.50%	+ 2.28%	+ 20.00%	+ 3.31%
NDCG@10	TIGER	0.0735	0.0099	0.0480	0.0923	0.0012	0.0563	0.0616	0.0029	0.0367
	Ours	0.0759	0.0142	0.0539	0.1020	0.0015	0.0650	0.0635	0.0035	0.0381
	$\Delta_{\mathcal{H}}$	+ 3.27%	+ 43.43%	+ 12.18%	+ 10.49%	+ 22.56%	+ 15.39%	+ 3.08%	+ 20.69%	+ 3.84%

Baselines. We compare our method, HypRQ-VAE with the following baseline methods, including standard sequential recommenders and LLM-based models: **MF** [33] decomposes the user-item interactions into the user embeddings and the item embeddings in the latent space. **Caser** [2] employs convolutional neural networks to capture users' spatial and positional information. **SASRec** [1] employs self-attention mechanisms to capture long-term dependencies in user interaction history. **P5-TID** [29] uses the title of items as textual identifiers for the LLM-based generative recommender model. **P5-CID** [29] incorporates collaborative signals into the identifier for LLM-based generative recommender models through a spectral clustering tree derived from item co-appearance graphs. **TIGER** [15] introduces codebook-based identifiers via RQ-VAE, which quantizes semantic information into code sequence for LLM-based generative recommendation. **LC-Rec** [16] uses codebook-based identifiers and auxiliary alignment tasks to better utilize knowledge in LLMs by connecting generated code sequences with natural language. **LETTER** [30] integrates hierarchical semantics, collaborative signals, and code assignment diversity to conduct effective item tokenization for LLM-based generative recommendation.

Implementation Details. To ensure a fair and reproducible comparison, we adopt the default hyperparameters from the official implementations of all baseline models. Evaluation is conducted using the leave-one-out strategy, where we perform a full ranking over the entire item set and report Hit Rate (Hit@K) and NDCG@K for $K \in \{5, 10\}$. For all generative models, the beam size is 20 during inference. Following prior work [16, 30], we apply a Trie-based constrained decoding [34], which restricts the autoregressive generation to valid successor tokens. Our proposed method is trained in two phases, as detailed in the following configurations. In Stage 1 (Semantic ID Generation), we utilize item titles and descriptions as input to generate hierarchical IDs. This stage is optimized using Adam with a learning rate of $1e{-}3$, employing a codebook architecture with $L = 4$ layers and $K = 256$ codes per layer. In Stage 2 (Generative Recommendation), we utilize a LLaMA2-7B backbone fine-tuned via LoRA ($r = 8, \alpha = 32, \text{dropout} = 0.05$). This phase uses the AdamW optimizer with a learning rate of $1e{-}4$ and a cosine schedule.

Prompt Template. Our approach follows an LLM-based generative framework, with next-item prediction as the primary training objective. Each instruction is personalized by combining a user's chronological interaction history, represented by the generated semantic IDs, with a natural-language prompt. To improve generalizability and robustness, we design a diverse set of prompt templates. One example is shown below:

Prompt Template

Instruction: The user has interacted with the following items in sequence: {Semantic IDs of interaction items}. Based on this interaction history, please predict the next item the user is most likely to engage with.

Response: {Semantic ID of target item}

B. Overall Performance

We first compare HypRQ-VAE against all baselines on three datasets, with the overall results reported in Table I. Several observations can be made. (1) Among LLM-based models with ID identifiers, P5-TID directly utilizes item titles as the identifier. Its relatively strongest performance on MovieLens suggests that longer token sequences may make accurate item prediction more challenging. (2) Models like P5-CID and LETTER benefit from incorporating collaborative signals, achieving strong performance on dense datasets such as MovieLens. (3) Our proposed HypRQ-VAE consistently outperforms all baselines in three datasets, highlighting the advantages of its underlying architecture. We attribute these gains to the

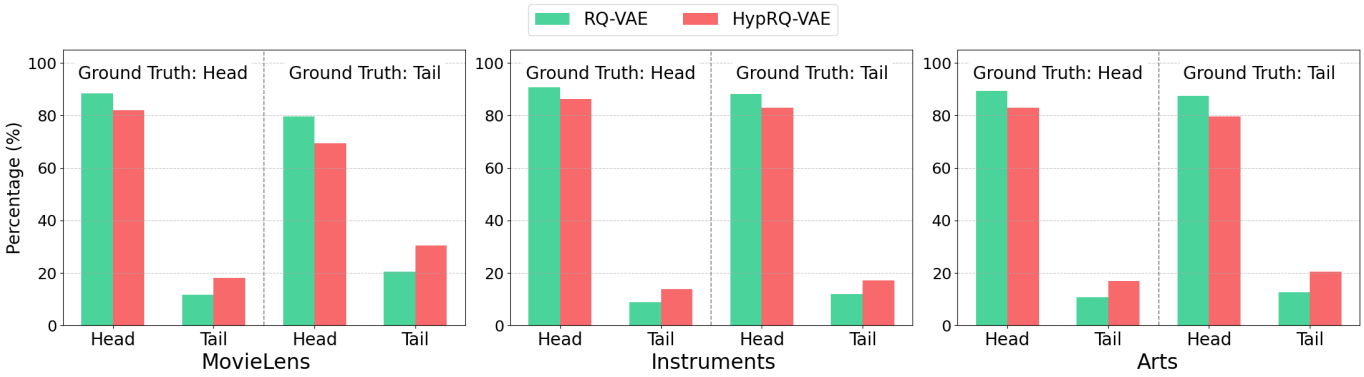


Fig. 5. Comparative performance of the RQ-VAE and HypRQ-VAE models regarding top-5 predictions across three datasets, highlighting each model's tendency to recommend popular (head) or niche (tail) items when evaluated against ground truth preferences.

exponentially expanding capacity of hyperbolic space, which provides greater representational flexibility for long-tail items than Euclidean embeddings and leads to improved overall recommendation performance.

C. Analysis of Long-Tail Performance

While hyperbolic models have shown promise, their specific advantages, especially for long-tail recommendation, remain underexplored. To address this, we investigate a critical question: *How does hyperbolic geometry specifically impact recommendation performance on head versus tail items?*

Performance on Head vs. Tail Items. To investigate the above question, we partition items in each dataset into head (top 20% most popular, H20) and tail (remaining 80%, T80) sets based on their popularity. We then compare the performance of HypRQ-VAE against its Euclidean counterpart, TIGER (RQ-VAE). The results in Table III reveal a clear pattern: while HypRQ-VAE improves performance on head items, the most substantial gains are consistently observed on tail items. For example, on MovieLens, HypRQ-VAE improves Hit@10 by over +52.71% for tail items, a much larger margin than for head items (+14.09%).

Furthermore, as shown in Figure 5, we analyze the composition of the top-5 recommendation lists (More detailed results, including top-10 recommendations, can be found in the Appendix). The results show that HypRQ-VAE recommends a significantly higher proportion of tail items than the Euclidean model, regardless of whether the ground truth item belongs to the head or tail. This confirms that our model does not just improve metrics but actively promotes item diversity.

Quantifying Representation Quality. We hypothesize that these performance gains stem from the ability of hyperbolic space to better represent sparse, long-tail items during quantization. In Euclidean space, distant and sparse tail items are often treated as outliers, leading to poor representations during quantization. In contrast, Hyperbolic space's exponential expansion and tree-like structure naturally accommodate this sparsity, allowing clustering centers to meaningfully represent both dense popular items and sparse niche items by preserving their structural relationships and maintaining appropriate distance during the quantization process. This geometry naturally organizes items into an implicit hierarchy. Popular items liked by many (e.g., flowers and fruit) are positioned near the hyperbolic origin—the "trunk" of the tree. Conversely, niche items appreciated by specific groups—like a paintbrush for a painter or a guitar for a musician—are placed further out towards the boundary, resembling the "branches" [19].

To test this hypothesis, we introduce the Quantization Error (QE) to measure the distance between an item's original embedding and its assigned cluster center (codebook vector). We calculate the Average Quantization Error (AQE) for both H20 and T80 in both spaces. Detailed results are shown in Table IV. While the absolute AQE values are not directly comparable due to their distinct geometric properties, the relative difference within each space is highly informative. The results reveal distinct patterns in how head and tail items are represented across different spaces. In Euclidean space, head items consistently have a lower AQE compared to tail items, confirming they are better represented. However, the hyperbolic space demonstrates a more balanced representation: the disparity between head and tail items is substantially reduced in the Instruments and Arts datasets. Most notably, on MovieLens, tail items achieve a lower AQE than head items, aligning with the massive performance boost observed for tail items in Table III. This provides strong quantitative evidence that hyperbolic geometry leads to a more balanced and effective representation for both popular and niche items.

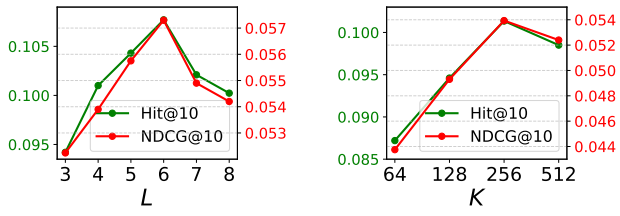


Fig. 6. Hyperparameters analysis of HypRQ-VAE on MovieLens.

D. Ablation Study

Length of Semantic IDs. We conducted an ablation study by varying the length L of the semantic IDs from 3 to 8 (see Figure 6). We can notice that performance rises steadily as L increases from 3 to 6, indicating that shorter identifiers may

TABLE IV
COMPARISON OF AVERAGE QUANTIZATION ERROR (AQE) FOR H20 AND T80 ITEMS. THE $\Delta_{\text{H-T}}$ COLUMN SHOWS THE PERCENTAGE DIFFERENCE BETWEEN H20 AND T80 AQE WITHIN EACH SPACE.

	# Items		AQE (Euclidean Space)			AQE (Hyperbolic Space)			Gain (%)
Dataset	H20	T80	H20	T80	$\Delta_{\text{H-T}}(\%)$	H20	T80	$\Delta_{\text{H-T}}(\%)$	
Movie Lens	551	2,200	0.025,64	0.025,68	-0.16	0.001,09	0.001,04	4.81	4.97
Instruments	1,985	7,937	0.022,91	0.023,67	-3.32	0.001,92	0.001,93	-0.52	2.80
Arts	4,192	16,764	0.021,78	0.022,76	-4.50	0.001,33	0.001,34	-0.75	3.75

fail to encode sufficient fine-grained details, thus limiting their expressiveness. However, further increasing L from 6 to 8 results in a performance decline. This is a plausible outcome of error accumulation inherent in autoregressive generation, as the task of accurately generating a longer sequence of codes is inherently more challenging than generating a shorter one, thus affecting the overall item prediction accuracy.

Codebook Dimension. We explored the effect of codebook dimension K on the performance of HypRQ-VAE by testing values of 64, 128, 256, and 512, as shown in Figure 6. The results suggest that a gradual increase in K generally leads to improved performance. The lower performance observed with smaller codebooks may be attributed to insufficient code diversity, limiting the model's capacity to differentiate items effectively. However, excessively large codebooks can lead to a decline in performance, potentially due to increased sensitivity to noise in the item's semantic information, which can result in overfitting on less meaningful aspects.

V. RELATED WORK

Generative Recommendation. Large Language Models (LLMs) enable generative recommendation by framing it as a sequence-to-sequence task. A key challenge is item indexing, which maps items into discrete tokens that LLMs can process. Existing approaches mainly use ID identifiers [29], textual identifiers [35], or codebook-based identifiers [15]. ID-based methods, such as SemID and CID [29], construct hierarchical structures from item semantics or co-occurrence graphs, but their fixed architectures may struggle to capture nuanced item relationships [36] and adapt to cold-start items or evolving catalogs. Textual identifiers use metadata such as titles and descriptions but produce long sequences and are prone to hallucination. Codebook-based methods instead learn discrete codes through multi-level vector quantization. TIGER [15] uses RQ-VAE to quantize semantic vectors, LC-Rec [16] applies Sinkhorn–Knopp to promote uniform code assignment, and LETTER [30] introduces regularization to improve code diversity and incorporate collaborative signals. OneRec [37, 38] generates hierarchical semantic IDs via RQ-KMeans, an iterative clustering algorithm that decomposes item embeddings into coarse-to-fine discrete tokens by performing balanced K-Means clustering on the residuals of each preceding layer. Building on multi-modal integration, MQL4GRecc [39] and MMQ [40] map information from diverse modalities into a unified latent language to facilitate knowledge transfer. Finally, to address the latency bottlenecks of autoregressive decoding, RPG [41] produces unordered, long semantic IDs, enabling the model to predict all tokens in parallel. Despite these advancements, existing codebook methods operate almost exclusively in Euclidean space, which struggles to model the hierarchical "power-law" structure inherent in recommendation data. In Euclidean manifolds, the limited representational capacity often leads to collisions and poor performance for long-tail items. In this work, we propose HypRQ-VAE a novel framework that generates semantic IDs within a hyperbolic manifold. By leveraging the exponential volume expansion of hyperbolic space, our model provides superior representational fidelity for niche items and naturally aligns with the hierarchical nature of item semantics.

Hyperbolic Learning. Growing evidence shows that many real-world datasets exhibit non-Euclidean structure and benefit from hyperbolic representations [42, 43]. Early work applied Poincaré embeddings to capture hierarchical relationships in complex networks [25–27]. More recently, hyperbolic geometry has been incorporated into advanced deep architectures, such as the recurrent neural network [44], attention network [45], and graph neural network [46] to exploit its ability to model hierarchy and hierarchy-like structure. In recommender systems, hyperbolic approaches such as HGCF [20] and HICF [19] have demonstrated superior preference modeling over Euclidean baselines. In contrast to these studies, which focus on traditional collaborative filtering, our work is the first to integrate hyperbolic embeddings into the generation of semantic IDs, opening new avenues for hyperbolic-driven generative recommendation.

VI. CONCLUSION

In this paper, we proposed HypRQ-VAE, the first framework to integrate hyperbolic geometry into residual quantized variational autoencoders for item indexing in generative recommender systems. By leveraging the exponentially expanding capacity of hyperbolic space, HypRQ-VAEbetter captures hierarchical item structure and improves the representation of long-tail items. Experiments on multiple benchmark datasets show that HypRQ-VAEconsistently outperforms Euclidean-based methods, with particularly strong gains on long-tail recommendation. Further analysis attributes these improvements to more balanced and effective quantization of both head and tail items in hyperbolic space. Overall, this work

demonstrates the effectiveness of hyperbolic geometry for generative recommendation and highlights its potential for more expressive and manifold-aware item indexing..

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APPENDIX

TABLE V
HEAD/TAIL COMPOSITION (%) OF TOP-k RECOMMENDATIONS.

Dataset	**Method**	GT_H-R_H	GT_H-R_T	GT_T-R_H	GT_T-R_T
Top-5 Predictions					
MovieLens	RQ-VAE	88.32%	11.67%	79.48%	20.51%
	HypRQ-VAE	81.88%	18.09%	69.43%	30.57%
Instruments	RQ-VAE	91.21%	8.79%	88.15%	11.85%
	HypRQ-VAE	86.18%	13.82%	82.82%	17.18%
Arts	RQ-VAE	89.29%	10.71%	87.41%	12.59%
	HypRQ-VAE	83.03%	16.97%	79.53%	20.47%
Top-10 Predictions					
MovieLens	RQ-VAE	86.26%	13.73%	77.82%	22.17%
	HypRQ-VAE	79.32%	20.65%	67.23%	32.75%
Instruments	RQ-VAE	88.65%	11.35%	85.24%	14.76%
	HypRQ-VAE	83.88%	16.12%	80.14%	19.86%
Arts	RQ-VAE	86.34%	13.66%	84.39%	15.61%
	HypRQ-VAE	79.96%	20.04%	76.68%	23.32%

Performance Analysis on Head vs. Tail Items. Table V compares the proportions of head and tailitems in the top-k recommendations, where GT_H/GT_T denotes a head/tail ground-truth item and R_H/R_T denotes the proportion of H20/T80 items in the predicted list.

The results consistently show that the HypRQ-VAE model predicts a higher proportion of tail items compared to the standard RQ-VAE. This is evident across all three datasets and for both top-5 and top-10 predictions, regardless of whether the ground truth item is a head or tail item. When the ground truth is a head item (GT_H), RQ-VAE's predictions are heavily skewed towards other head items, exhibiting a high R_H. In contrast, HypRQ-VAE demonstrates a lower R_H and a higher R_T in its predictions compared to RQ-VAE. This indicates that HypRQ-VAE is more effective at incorporating a mix of tail items. When the ground truth is a tail item (GT_T), the difference is even more pronounced. HypRQ-VAE significantly increases the proportion of tail items in its predictions, with its R_T values being substantially higher than those of RQ-VAE. This demonstrates HypRQ-VAE's enhanced ability to recommend less popular, long-tail items, which is particularly beneficial for long-tail recommendation scenarios.

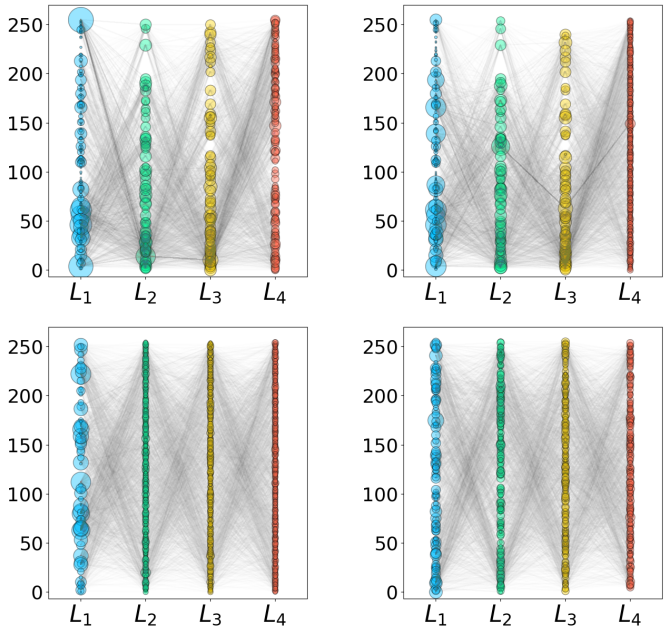


Fig. 7. Code paths across codebook layers on MovieLens for TIGER (top left), LC-Rec (top right), LETTER (bottom left), and HypRQ-VAE (bottom right). HypRQ-VAE achieves a more balanced token distribution across layers without additional regularization.

Visualization of Codebook Utilization. We further visualize the code path across codebook layers for different methods. The number of nodes for each layer matches the size of the corresponding codebook, and the node size reflects the degree of the respective token. The results are shown in Figure 7, and we can notice that: When compared to TIGER, LC-Rec shows a more uniformly distributed token assignment in its last layer, and LETTER achieves a more balanced distribution across its last three layers. Notably, HypRQ-VAE as a hyperbolic adaptation of RQ-VAE, achieves a remarkably even token distribution without the need for additional regularizers or constraints.

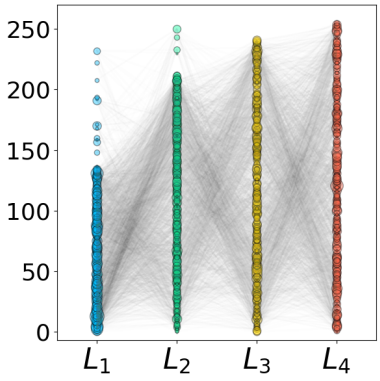


Fig. 8. Code distribution of HypRQ-VAE under uniform initialization.

Moreover, we investigated different initialization strategies and discovered an interesting phenomenon: when employing uniform distribution initialization, the code identifiers for each layer exhibited a distinctive distribution pattern (as shown in Figure 8). Furthermore, as the number of layers increased, additional representational space became necessary to differentiate increasingly granular levels of information. This observation suggests a promising direction for future work, specifically, the adaptive adjustment of dimensional capacity at each layer to optimize spatial utilization throughout the network architecture. Specifically, the previous layers might focus on broad semantic categories, requiring less space, while subsequent layers refine these into finer levels of information, demanding increased dimensionality. An adaptive approach could dynamically allocate resources, potentially leading to more compact and efficient representations.